

Taxing Income when Income Risk is Endogenous

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A workhorse taxation model (Mirrlees 1971):

- Government set a (nonlinear) income tax $T(z)$.
- Agent θ chooses labour ℓ , earns $z = \theta\ell$ and consumes $c = z - T(z)$.
- Agents control only the level (mean) of earnings.

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Q1: What are the incentives to manipulate risk?

Q2: What are the implications for the design of tax-and-transfer systems?

Manipulating earnings dispersion—possible channels:

1. **Explicit risk taking: self-employed and business owners, executives, portfolio managers** Chevalier and Ellison (1997); Cullen and Gordon (2007); Vereshchagina and Hopenhayn (2009)...
2. **Variable pay (bonus, commission, piece rate): common among the U.S. employees** Lemieux, MacLeod, and Parent (2009); Grigsby, Hurst, and Yildirmaz (2021)
3. **Shifting income across time: evidence for business owners and for employees:** Le Maire and Schjerning (2013); Miller, Pope, and Smith (2024); Kreiner, Leth-Petersen, and Skov (2014a,b)...
4. **More generally, many decisions affect income risk: education, occupation, financial investments**

Framework

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A tax-and-transfer schedule based on realised earnings z is $T(z)$.

The implied consumption function (after-tax income) is $C(z) := z - T(z)$.

The agent solves

$$U(\theta) = \sup_{\ell, \mu} \int_0^\infty u(C(z)) d\mu(z) - g(\ell) \quad \text{s.t.} \quad \int_0^\infty z d\mu(z) = \theta \ell \quad (P_\theta)$$

where

- μ is a probability measure (on the Borel space $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$),
- $C(z)$ is the consumption at earnings z ,
- the constraint ensures that mean earnings are equal $\theta \ell$.

Incentives to manipulate risk

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Proposition 1

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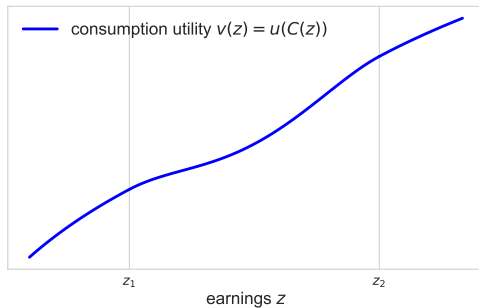
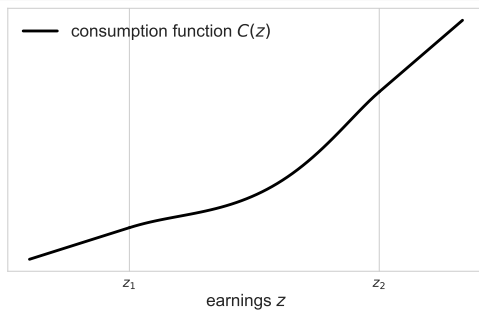
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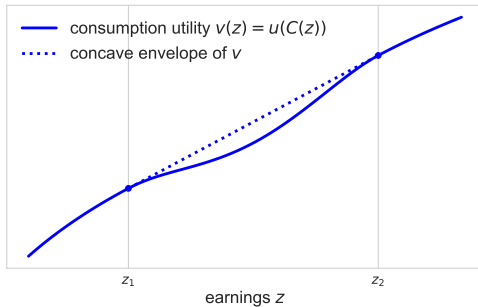
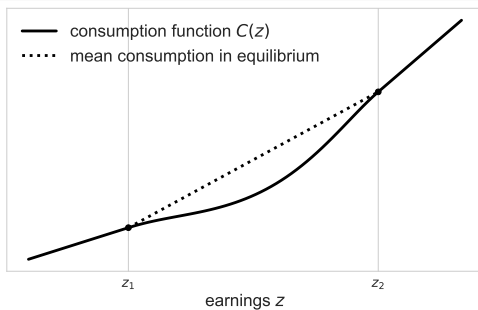
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Take-away: there are incentives for extra risk when $C(z)$ is "convex enough" to make $u(C(z))$ non-concave.

Example: $u(c) = \log c$



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Let $\gamma(z)$ be the coefficient of absolute risk aversion.

Concavity of $v(z)$ translates (under differentiability) into

$$T''(z) \geq -\gamma(z) \cdot (1 - T'(z))^2$$

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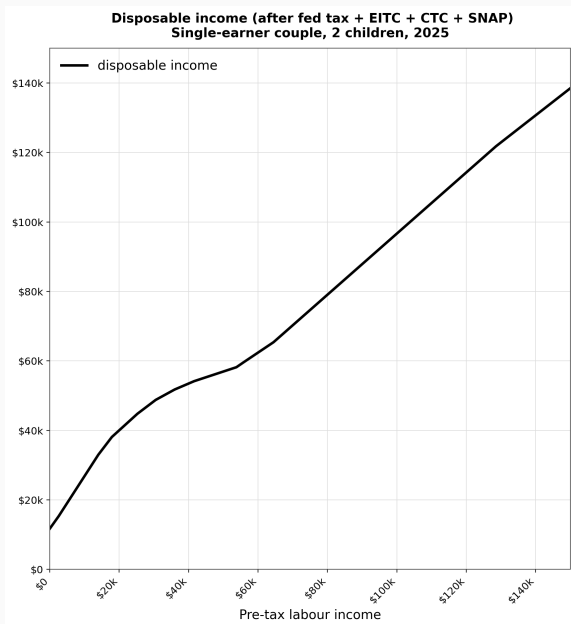
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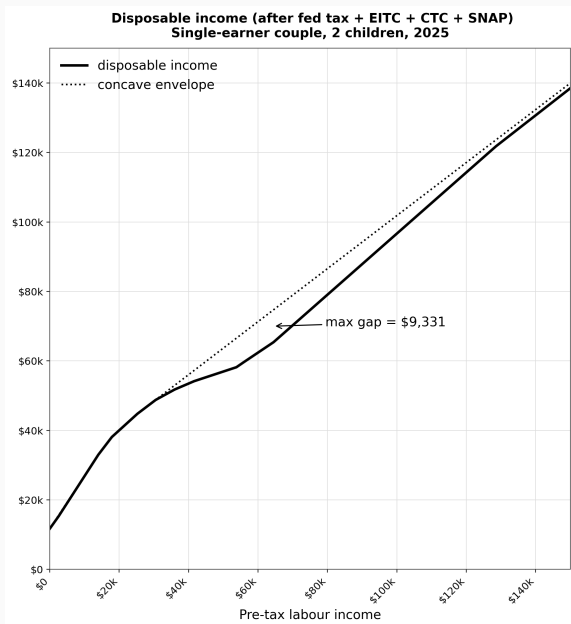
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Empirically, marginal tax rates often decline at incomes where the welfare benefits and tax credits are fully phased-out.

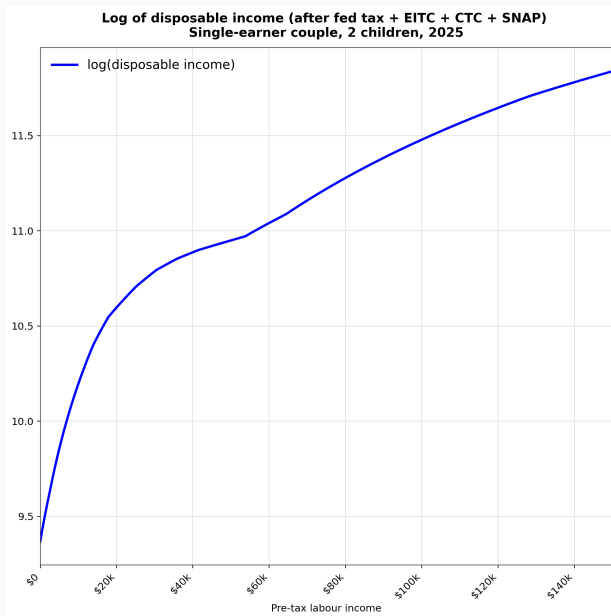
Tax-and-transfer schedule in the U.S.



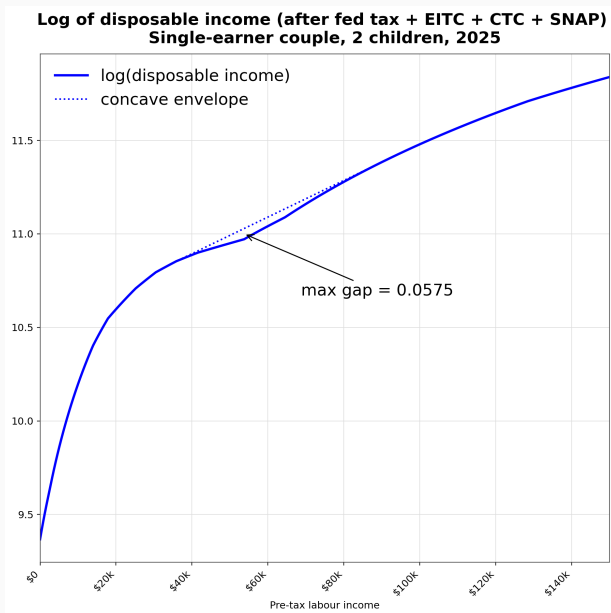
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Implications for the tax design

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Proof idea: Define C_1 as $u(C_1(z)) = \widehat{v}_0(z), \forall z$.

Let the planner have a weighted utilitarian objective, with weights $\omega(\theta)$.

Let T be the optimal tax schedule and $\gamma(z)$ the coeff. of absolute risk aversion.

Proposition 3

At any income level z where $v''(z) < 0$, the optimal tax schedule follows the Diamond-Saez formula:

$$\frac{T'(z)}{1 - T'(z)} = \frac{1}{\varepsilon_{z,1-\tau}(z)} \frac{1}{zh(z)} \int_z^\infty (1 - m(y)) e^{\int_z^y (1 - T'(x)) \gamma(x) dx} dH(y).$$

Elsewhere, the optimal tax schedule satisfies $v''(z) = 0$, or

$$T''(z) = \gamma(z)(1 - T'(z))^2.$$

- Assumed T continuously diff. and piecewise twice-diff.—rules out bunching.
- Notation: the cdf and density of earnings $H(z), h(z)$; the elasticity of earnings $\varepsilon_{z,1-\tau}(z)$; the marginal social welfare weight $m(z(\theta)) \propto \omega(\theta)u'(c(\theta))$.

Extensions:

- Multiple time periods and stochastic shocks
- Alternative dynamic interpretation: intertemporal shifting of income
- Extra frictions (e.g., moral hazard)
- Costly risk manipulation

Assumptions on preferences:

- Common risk preferences: important and cannot be easily relaxed.
- Common labour disutility: can be relaxed to (multidimensional) heterogeneity.
- Additive separability: can allow $u(c)d(\ell)$, but unlikely beyond that.

Quantitative exercise

Broad idea: fix preferences and find productivity distribution that replicates the observed labour income distribution.

Calibration details follow Heathcote and Tsujiyama (2021), with some modifications.

- Isoelastic preferences: $U(c, \ell) = \frac{c^{1-\sigma}}{1-\sigma} - \frac{\varepsilon^F}{1+\varepsilon^F} \ell^{1+1/\varepsilon^F}$, $\varepsilon^F = 0.5, \sigma = 1$.
- Empirical tax progressivity (HSV): 0.181; gov spending to GDP: 18.8%.
- Household income data: SCF 2007 (no censoring of low incomes).
- Income distributed as **Double Pareto-Lognormal**:

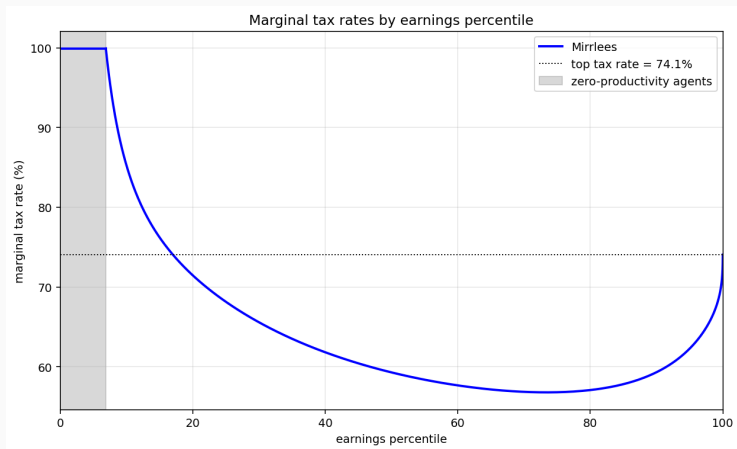
$$\log(z) = \mu + \sigma N + E_1/\alpha - E_2/\beta,$$

- $N \sim N(0, 1)$, $E_i \sim \text{Exp}(1)$, independent
- MLE tails coefficients: $\alpha = 2.1, \beta = 1.4$ ▶ tail figure ▶ Robustness check (CPS)
- **6.9% of households with no labour income → assign 0 productivity.**

I compare the optimal tax-and-transfer schedule under two scenarios:

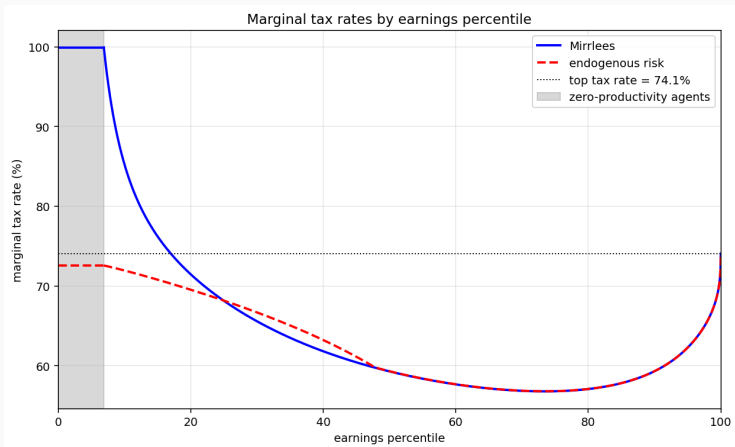
1. "Mirrlees": risk manipulation is exogenously ruled out;
2. "endogenous risk": risk manipulation is possible, but optimally prevented.

Utilitarian planner, $u(c) = \log c$



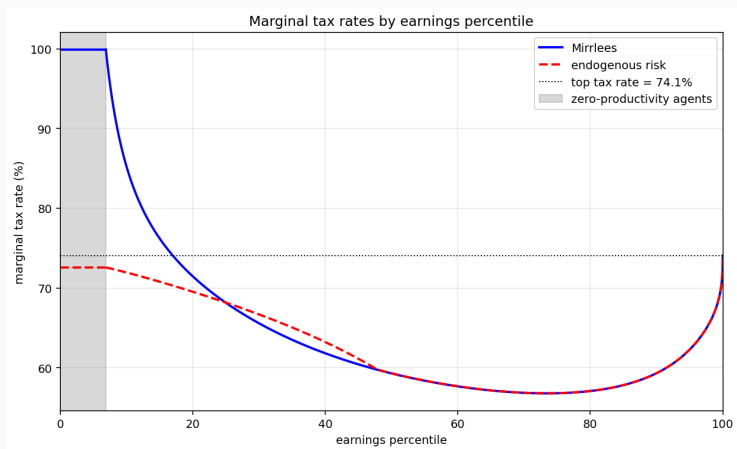
- Mirrlees: high and strongly declining MTRs at the bottom $\rightarrow$ high transfer that is quickly phased out.

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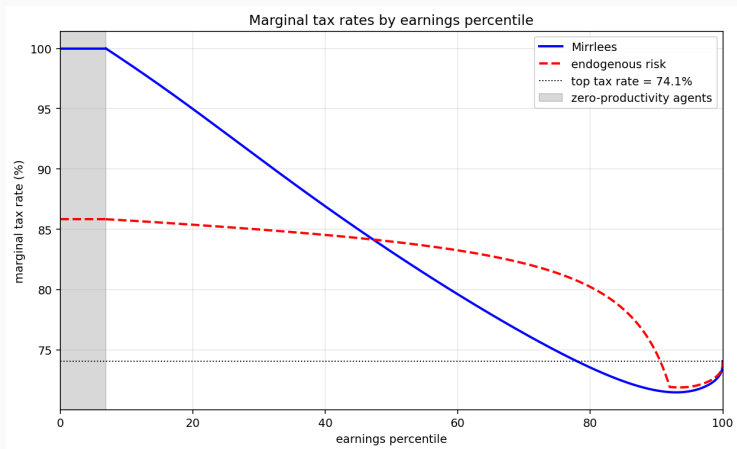
- Endogenous risk: MTR at the bottom is lower (by 27 pp) and flatter → lower transfer that is phased out more slowly

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- Endogenous risk: MTR at the bottom is lower (by 27 pp) and flatter → lower transfer that is phased out more slowly
- Welfare change (Mirrlees → endogenous risk): -0.046% CEV.

Rawlsian planner, $u(c) = \log c$

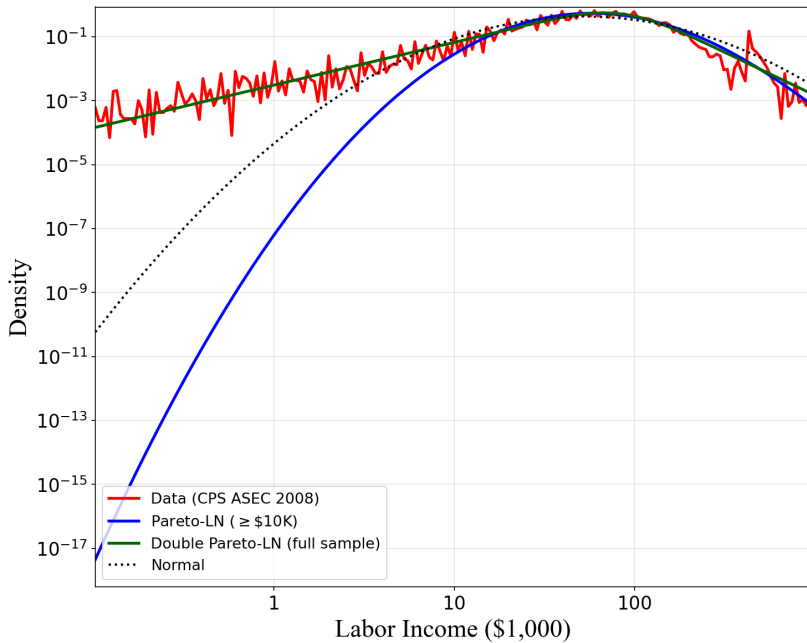


- Mirrlees: MTRs decrease with income over most of the distribution.
- With risk manipulation, the schedule of MTRs substantially flattened.
- Welfare change (Mirrlees $\rightarrow$ endogenous risk): -2.24% CEV.

Conclusions

- Theory and quantitative exploration of incentives to manipulate earnings risk and implications for tax design.
- When agents control the dispersion of earnings, decreasing marginal tax rates give incentives for inefficient risk manipulation.
- Marginal tax rates are often decreasing at income levels where the social transfers are phased-out.
- Removing incentives for risk manipulation leads to smaller transfers that are phased-out more slowly.

Extra figures



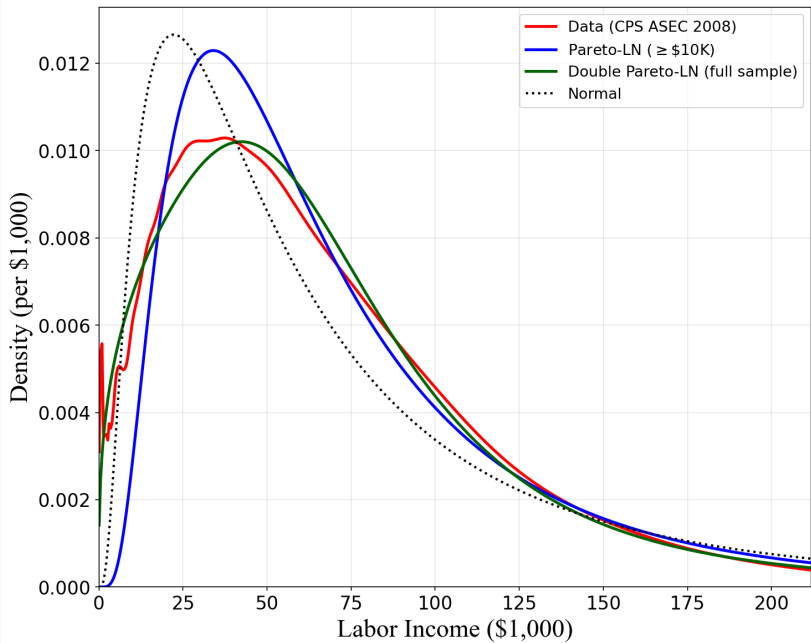


Table 1: Difference in marginal tax rate at the bottom (relative to Mirrlees, pp)

	$\sigma = 0$	$\sigma = 0.5$	$\sigma = 1$	$\sigma = 2$
utilitarian	0	-45	-27	-1
utilitarian (weighted ¹)	-62	-33	-21	-1
Rawlsian	-33	-22	-14	-1

- Largest impact when risk aversion is low.
- Under risk neutrality, optimal tax schedules that prevent risk manipulation are fully linear.

◀ Return

¹Welfare weights follow $\omega(\theta) \propto F(\theta)^{-\frac{1}{2}}$.

Table 2: Welfare loss due to risk manipulation (relative to Mirrlees, CEV)

	$\sigma = 0$	$\sigma = 0.5$	$\sigma = 1$	$\sigma = 2$
utilitarian	0%	-0.05%	-0.05%	-0.002%
utilitarian (weighted ²)	-1.3%	-0.5%	-0.3%	-0.1%
Rawlsian	-9.9%	-4.1%	-2.2%	-0.9%

- Welfare loss is increasing in the strength of redistributive preferences and decreasing in risk aversion.

[← Return](#)

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